

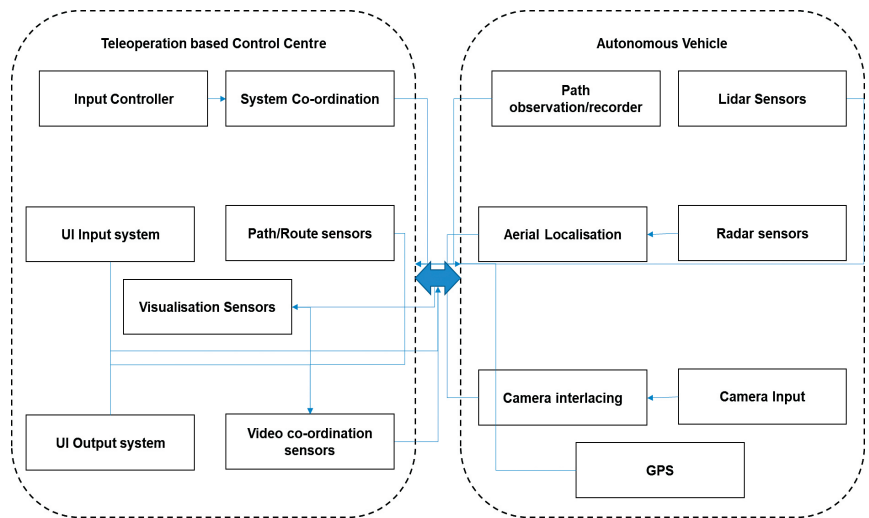
keeping. In this case, driver can take hands off the vehicle and relax, but still needs to keep a close eye on the surrounding environment. A new trend under level 2 is Level 2 + Systems, where high degree of automation is observed while entering and existing a highway or lane. Examples: GM's Super Cruise, Mercedes-Benz Drive Pilot, Tesla Autopilot, Volvo's Pilot Assistance, and Nissan ProPilot Assist 2.0

Level 3: Conditional Automation. Under conditional automation, the car takes over most of the controls of driving including environmental monitoring. The driver has to pay attention and be ready to take over the control when vehicle prompts. Level 3 automation assists the driver to take off the hands and control from car as long as the driver remains alert. The speed limit where Level 3 conditional automation is recommended is up to 65 kilometres per hour. Example: Audi Traffic Jam Pilot.

Level 4: High Automation. Under this automation, the car can operate without any sort of human assistance but only under selected conditions defined by factors like type of road or geographical area. All the major functionalities like steering, braking, accelerating, and environmental monitoring including lane changing, turning, and signalling can be done by car's onboard computer. Example: Google's now-defunct Firefly pod-car prototype, which is restricted to the speed of 40kmph.

Level 5: Full Automation. Level 5 autonomy requires zero human attention. There's no need for a steering wheel, no need for brakes, and no need for pedals. The autonomous vehicle controls all driving tasks under all conditions, including the monitoring of environment and identification of complex driving conditions like busy pedestrian crossings.

This also means that the vehicle can perform a combination of several tasks simultaneously, whether adaptive cruise control, traffic sign recognition, lane departure warning, emergency braking, pedestrian detection, collision avoidance, cross traffic alert, surround view, park assist, rear collision warning, or park



Reference architecture of teleoperations based autonomous vehicle system

assistance. Example: Waymo project that has yet to design Level 5 tech for self-driving.

Teleoperations and sensors

As per <https://www.sae.org/automated-unmanned-vehicles/vehicle> automation is classified into five levels (actually six levels in which Level 0 is fully manual driver operation and hence ignoring it). Autonomous vehicle and autopilot application is a complex and contentious technology as it involves a large amount of internal processing using data collected from various sensors and remote operation from outside the line of sight called Teleoperations.

Some of the teleoperations facilities are there for quite some time like anti-lock braking or electronic stability control (ESC), which are used in semi-automatic cars and then have slowly emerged to advanced sensor operations like advanced driver assistance system (ADAS) and crash avoidance system (CAS) which acts quickly during accident times (such as driver protection and alarm sensors). Teleoperation is based on three major controls of autonomous vehicle and its autopilot application design. These are:

Acceleration. To handle the vehicle steering with throttling speed variations based on inputs from obstacles (speed breaker, signage) and angle changes in route.

Direction. Steering action control system to handle left and right move-

ment of the vehicle based on route change and lidar, radar sensor inputs.

Stopping. Brake handling control system to control acceleration and stop the vehicle based on input from different sensors including GPS and IMU.

Teleoperation uses large volume of data and hence does fast processing of data and gives quick commands to control the sensors. Latest advancements like 360° camera view can help in enhancing the autopilot applications for better sensor inputs for vehicle movement and teleoperation activities and control.

So, autopilot application is a revolutionary solution in the automobile industry to create semi-automated and fully-automated vehicles and enable various facilities for physically challenged or visually impaired drivers to control vehicles using voice commands. **EFY**

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